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Synthesis and Performance Evaluation of a Multifunctional High-Performance Drag Reducer

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Abstract

Low-viscosity slickwater fracturing fluids have been extensively employed in the exploration and development of shallow shale gas reservoirs in the Gulong region. Conventional slickwater systems fail to meet field operational requirements as exploration advances into deep, high-pressure shale formations. Consequently, there is an urgent need to develop a novel fracturing fluid capable of real-time switching between low-viscosity and high-viscosity fluids to reduce operational costs. In this study, a novel friction reducer (PLD) was synthesized using acrylamide (AM), acrylic acid (AA), 2-acrylamido-2-methylpropanesulfonic acid (AMPS), and a dual-hydrophobic long-chain monomer (LD). A series of laboratory tests were conducted to evaluate the system's rheological properties, friction reduction performance, salt tolerance, and proppant-carrying capacity. Rheological tests demonstrated a significant increase in viscosity as the PLD concentration raised from 0.1 wt% to 2.0 wt%. The low-concentration PLD solution exhibited outstanding friction reduction, achieving a rate exceeding 75% across a wide flow velocity range. High-concentration PLD demonstrated good salt tolerance, shear resistance, and proppant-carrying capacity. At a salinity of 20,000 mg/L, PLD maintained good stability. Temperature and shear resistance tests showed that the 1.2 wt% PLD solution at 120 °C and the 2.0 wt% PLD solution at 145 °C both maintained a final apparent viscosity above 50 mPa·s. Proppant-carrying tests confirmed its effective proppant suspension capability. In summary, the PLD friction reducer exhibits high-efficiency friction reduction at low concentrations and significant proppant-carrying capacity at high concentrations, making it suitable for hydraulic fracturing operations in complex-geology shale gas reservoirs.

Introduction

The Gulong shale oil reservoir is characterized by well-developed laminae fractures. The reservoir exhibits small pore sizes and narrow throats, which result in extremely low permeability (Zhang et al., 2023; He et al., 2022). Horizontal permeability is mainly influenced by the development of bedding fractures, while vertical

permeability is determined by a combination of laminae fractures and the matrix. The difference between horizontal and vertical permeability can be up to several orders of magnitude (Ju et al., 2023; Yu et al., 2022). The geological properties of the Gulong shale, such as extensive laminae fractures, limited vertical fracture propagation during stimulation, and significant disparity between vertical and horizontal flow capacity, present major challenges for oil and gas development (Sun et al., 2021). Hydraulic fracturing operations can generate complex artificial fracture networks that effectively interconnect natural fracture systems and significantly enhance overall reservoir permeability. This enables efficient extraction of otherwise challenging shale oil resources. Therefore, large-scale hydraulic fracturing and other stimulation measures are necessary to improve reservoir flow conditions (Lei et al., 2025; Meng et al., 2024; Liu et al., 2023).

Common fracturing fluids primarily include guar gum fracturing fluids, cellulose-based fracturing fluids, and polymer fracturing fluids, each with distinct characteristics in shale oil reservoir stimulation (Bai et al., 2025; You et al., 2025). Although guar gum fracturing fluids exhibit excellent thickening performance, they leave a high residue content after gel breaking, which can easily clog the reservoir's micro-pores, significantly damaging permeability and impairing long-term productivity. Cellulose-based fracturing fluids, while cost-effective, suffer from poor chemical stability and inadequate resistance to acids and alkalis, making them prone to degradation under complex formation conditions and compromising fracturing effectiveness. In contrast, polymer fracturing fluids, leveraging their molecular structure advantages, demonstrate superior water solubility, high-efficiency thickening capability, and outstanding temperature and shear resistance. They can adapt to high-temperature, deep-well operational environments while overcoming the drawbacks of traditional plant gum and cellulose fracturing fluids, such as excessive residue and poor stability, making them a crucial option for current shale oil reservoir stimulation (Zhang et al., 2025).

Synthetic polymer fracturing fluids are high-performance fracturing fluid systems formulated using polyacrylamide, polyvinylamine, or polyacrylamide derivatives as the primary thickeners, combined with specific inorganic or organic crosslinking agents (Sun et al., 2014). Under the action of crosslinkers, these fracturing fluids can form a stable three-dimensional spatial network structure, exhibiting unique viscoelastic properties. Their key advantage lies in the flexible adjustment of formulations based on actual needs—by modifying parameters such as polymer molecular weight, crosslinker type, and dosage, precise control over the system's rheological properties and gel-breaking characteristics can be achieved (Yang et al., 2023). Compared to conventional fracturing fluids, synthetic polymer fracturing fluids offer significant performance advantages, including lower residue content, superior temperature and shear resistance, and controllable gel-breaking behavior (Zuo et al., 2025; Zhao et al., 2024; Huang et al., 2024). These characteristics make them particularly effective in high-temperature, deep reservoir stimulation, where they can effectively prop fractures while minimizing secondary damage to the reservoir. As such, they are especially well-suited for hydraulic fracturing operations in low-permeability oil and gas reservoirs.

Hydrophobically associating polymers form dynamic physical crosslinking networks through the aggregation of hydrophobic groups on their molecular chains. This unique associative interaction can significantly alter fluid rheological properties, enhance solution viscoelasticity, and impart excellent thickening capability and shear recovery characteristics to fracturing fluids. Based on colloid chemistry and polymer chemistry theories, Cao et al. (2008) systematically investigated the molecular design, synthesis methods, performance regulation, and industrial applications of water-soluble hydrophobically associating polymers. They developed high-performance associating polymer products with superior thickening properties, which have been successfully applied in oilfield development. Xiao et al. (2009) synthesized a novel polymeric thickener by copolymerizing AM, salt-resistant monomers, and hydrophobic monomers. This system forms a gel fracturing fluid with outstanding temperature and salt resistance through crosslinking between polynuclear hydroxo-bridged titanium complex ions in acidic crosslinkers and the amide groups of the polymer. Wu et al. (2016) successfully prepared a temperature- and salt-resistant quadripolymer hydrophobically associating polymer, AAMS-2, via inverse emulsion

polymerization using four monomers: AM, AMPS, dimethyloctadecyl allyl ammonium chloride, and sodium 4-acrylamidobenzenesulfonate. [Deng et al. \(2021\)](#) synthesized a high-performance associating polymer, DSAM, by copolymerizing AM, AMPS, AA, and hydrophobic monomer AMD-12. The resulting polymer exhibits excellent temperature resistance (up to 120°C) and outstanding shear stability.

The Gulong shale reservoir exhibits certain unique characteristics. In recent years, experts and scholars have developed various fracturing fluids tailored to this formation based on its specific properties. [Wu et al. \(2022\)](#) innovatively designed a novel thickener molecule containing bulky rigid groups by combining microemulsion phase behavior theory and molecular structure-activity relationships, specifically targeting the characteristics of the Gulong shale reservoir and its crude oil composition. This unique structure significantly enhances the temperature and shear resistance of the fracturing fluid. To meet the stimulation needs of low-permeability reservoirs in the periphery of the Daqing Oilfield, [Li \(2011\)](#) employed the hydrophobically associating water-soluble polymer JR200 as a thickener for fracturing fluids. Field tests demonstrated favorable production enhancement effects in low-permeability reservoirs. [Liu et al. \(2023\)](#) analyzed the impact of different fracturing fluids on the Gulong shale reservoir. Their study evaluated the adsorption damage caused by polymer fracturing fluids, slickwater, weakly alkaline guar gum fracturing fluids, and acidic fracturing fluids on laminated, bedded, and massive shale reservoirs. The results suggested that slickwater is more suitable for the Daqing shale reservoir. Considering the well-developed bedding fractures and natural fractures in the Gulong shale reservoir, [Zhang et al. \(2020\)](#) found that high-viscosity fracturing fluids facilitate fracture height growth, while low-viscosity fluids promote the opening of bedding fractures. Field tests showed that combining gel to enhance fracture height and injecting slickwater to connect bedding fractures can effectively improve shale reservoir stimulation. Through extensive research by numerous experts, the Gulong shale fracturing fluid system has been progressively refined. However, several challenges remain: conventional fracturing fluids cause significant residue damage, hydrophobically associating polymers exhibit insufficient temperature resistance, their salt tolerance requires further improvement, and there is a lack of high-efficiency friction-reducing systems specifically designed for the nanoscale pore-throat characteristics of this reservoir.

In response to the shortcomings of current research, this study aims to optimize the performance of hydrophobically associating polymers according to the geological conditions of the Gulong formation. A multifunctional high-efficiency friction reducer, PLD, was synthesized using AM, AMPS, AA, and a structurally tailored hydrophobic monomer LD. At low concentrations, PLD demonstrates excellent friction-reducing performance and can form a slickwater system when combined with flowback aids and swelling agents. Without the addition of crosslinkers, high-concentration PLD can serve as a thickener for fracturing fluids, exhibiting superior viscoelasticity and proppant-carrying capacity. This enables it to meet the diverse operational requirements of low-permeability shale formations like Gulong under varying field conditions.

Materials, Equipment, and Workflow

Materials

Acrylamide (AM), acrylic acid (AA), 2-acrylamido-2-methylpropanesulfonic acid (AMPS), sodium hydroxide, potassium persulfate (KPS), sodium bisulfite, and potassium chloride, all of analytical reagent grade, were supplied by Beijing InnoChem Science & Technology Co., Ltd. The polymerizable hydrophobic long-chain monomer LD was synthesized in our laboratory. Hydroxypropyl guar gum fracturing fluid was commercially sourced.

Equipment

HAAKE MARS 60 high-temperature, high-pressure rheometer (Thermo Fisher Scientific Inc.); HTP-DS high-temperature, high-pressure core displacement apparatus (Jiangsu Lianyou Scientific Instrument Co., Ltd.); fracturing fluid pipeline drag reduction tester (China University of Petroleum, Beijing).

Synthesis of the Drag Reducer

A compound emulsifier was added to a three-necked flask containing white oil. The system was purged with nitrogen for deoxygenation and stirred mechanically for 20 min. Acrylamide (AM), acrylic acid (AA), and the laboratory-synthesized hydrophobic long-chain monomer LD were dissolved in deionized water under stirring until completely dissolved. The pH was then adjusted to approximately 7 using sodium hydroxide. The aqueous solution was added dropwise to the oil phase through a constant-pressure dropping funnel under a continuous nitrogen atmosphere while controlling the stirring speed. After heating to approximately 35 °C using an oil bath, an appropriate amount of initiator was added. The reaction was carried out at this temperature for 6 hours. The mixture was then cooled to room temperature, and a phase transfer agent was added under stirring to obtain a milky-white PLD emulsion.

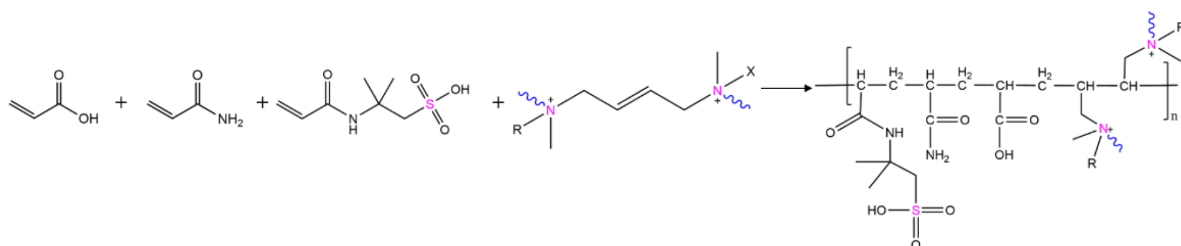


Figure 1—Schematic diagram of the PLD molecular structure.

Results and Discussion

Solubility and Viscosity Evolution

PLD solutions at varying concentrations were prepared to assess solubility. Viscosity measurements were conducted using a HAAKE MARS 60 rheometer equipped with a CC41 coaxial cylinder rotor system under a constant shear rate (170 s^{-1}) at 25 °C for 10 min. Fig. 2 illustrates the dissolution process of PLD in water, demonstrating rapid viscosity development within 30 seconds and near-complete dissolution within 10 minutes, confirming its excellent solubility. As shown in Fig. 3, the PLD system exhibits low viscosity at dilute concentrations, while a significant viscosity increase occurs with rising concentration. At concentrations exceeding 1.2 wt%, viscosity surpasses 100 mPa·s. Below the critical micelle concentration (CMC) of PLD, intramolecular interactions dominate among hydrophobic groups, resulting in only a marginal viscosity enhancement. At elevated concentrations, increased molecular density promotes intermolecular association of hydrophobic moieties. This intermolecular association facilitates the entanglement of the dual hydrophobic chains, expands the hydrodynamic volume, and establishes network structures, all of which collectively enhance the thickening efficacy.

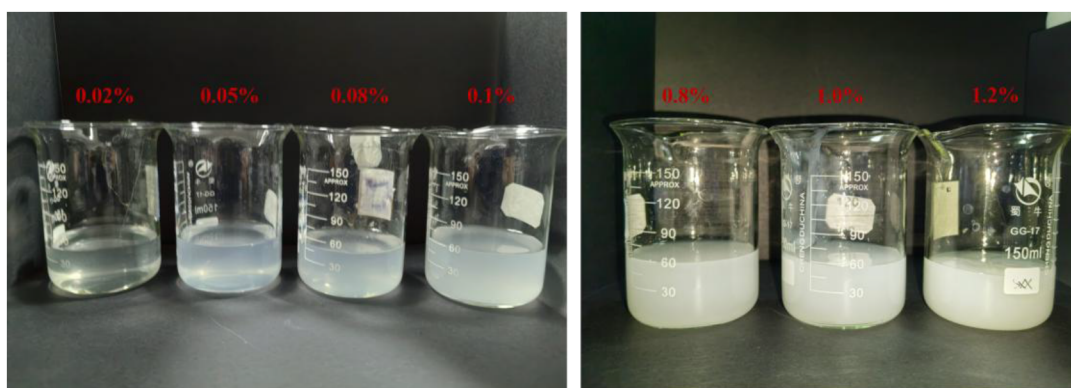


Figure 2—Dissolution behavior of PLD aqueous solutions at varying concentrations.

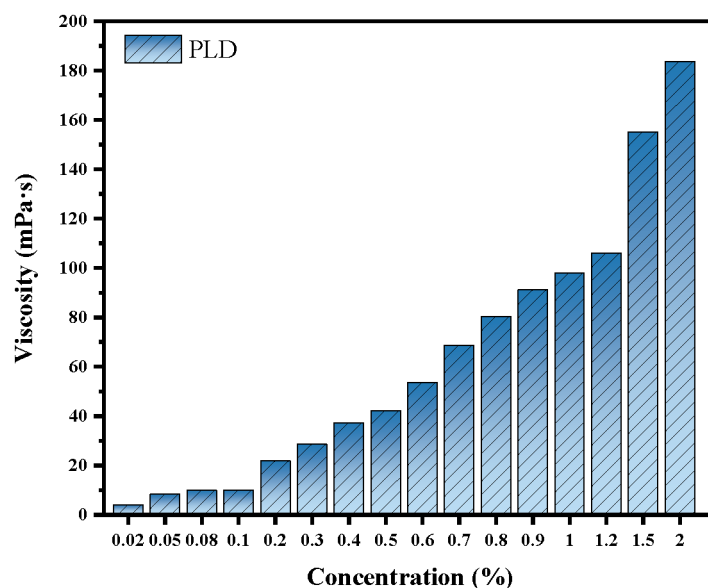


Figure 3—Viscosity evolution of PLD aqueous solutions with increasing concentration.

Drag Reduction Performance

Frictional pressure drops of water and PLD solutions at varying concentrations were measured using a fracturing fluid pipeline drag reduction tester under identical flow velocities. The drag reduction ratio of PLD was calculated from these measurements. As depicted in Fig. 4(a), at a flow velocity of 12 m/s, the drag reduction ratio increases rapidly with PLD concentration in the 0.03-0.3 wt% range before reaching a plateau. Both 0.1 wt% and 0.3 wt% solutions exceed a 75% drag reduction ratio. This is because, below 0.1 wt%, an insufficient number of molecular chains interact with the pipe wall, resulting in limited drag reduction. At 0.1 wt%, enhanced intermolecular interactions significantly improve drag reduction efficiency. No significant change in the drag reduction rate is observed with a further increase in concentration.

Fig. 4(b) demonstrates the flow velocity dependence of the drag reduction. At fixed concentrations, the drag reduction ratio initially rises with velocity before stabilizing. Under low-flow conditions (laminar flow), disordered molecular alignment results in modest effectiveness. At high velocities (turbulent flow), the hydrophobic chains adopt stretched, linear conformations, dramatically enhancing performance. Notably, the 0.1 wt% PLD solution achieves a 70% drag reduction at a flow rate of merely 5 m/s. This property can be attributed to the influence of its unique molecular architecture, which features a dual long-chain structure.

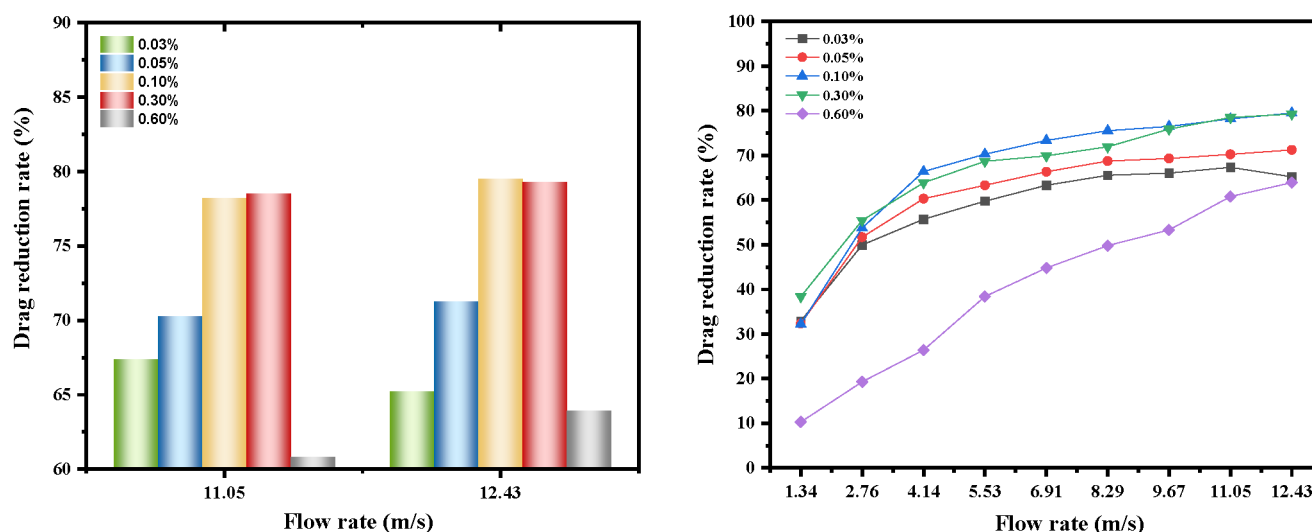


Figure 4—(a) Drag reduction efficiency of PLD at high shear rates;
(b) Concentration- and velocity-dependent drag reduction behavior.

Effect of Shear Rate

To investigate shear-induced structural changes in PLD, viscosity evolution was measured across shear rates of 1-1000 s^{-1} . As shown in Fig. 5, PLD solutions exhibit marked shear-thinning characteristics, with viscosity decreasing sharply at low shear rates. Below low shear rate, highly entangled polymer chains adopt random coil conformations resulting in elevated viscosity due to significant interchain friction resistance. With increasing shear rate, chain disentanglement and molecular rearrangement occur, substantially reducing viscosity. Above 500 s^{-1} , viscosity plateaus to form a distinct second Newtonian region, attributable to the dual hydrophobic chains in PLD's molecular architecture conferring exceptional shear resistance. Under high shear, dynamic equilibrium establishes between chain disentanglement and flow-induced orientation, stabilizing viscosity. These findings demonstrate PLD's superior shear-recovery capability.

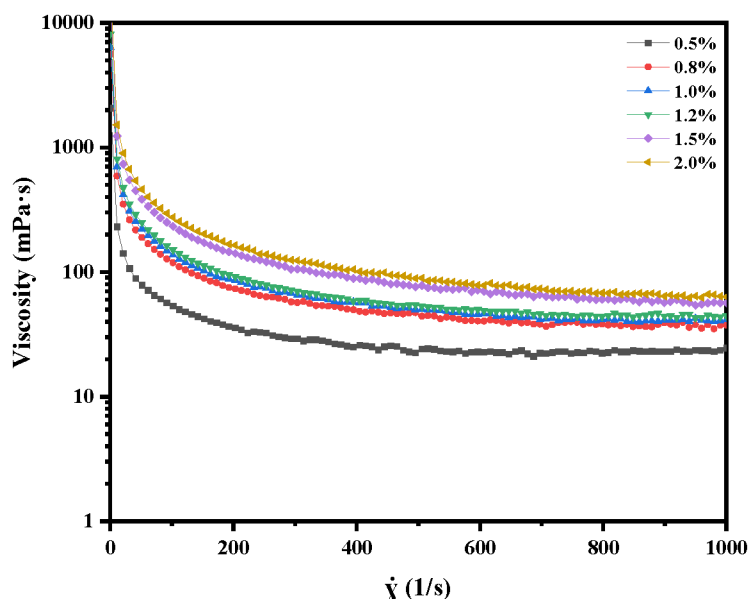


Figure 5—Shear-thinning behavior of PLD solutions at varying concentrations.

Viscoelastic Characterization

Dynamic rheological tests were conducted at 25 °C. Amplitude sweeps (0.01–10 Pa shear stress at 1.0 Hz) first determined the linear viscoelastic region (LVE). Subsequent frequency sweeps (0.01–10 Hz) at fixed 1 Pa shear stress (within LVE) evaluated viscoelastic properties of PLD solutions. As shown in Fig. 6, the storage modulus (G') consistently exceeded the loss modulus (G'') across 0.01–10 Hz. With increasing PLD concentration (0.1–1.2 wt%), both moduli increased, though G' demonstrated substantially greater enhancement than G'' . This phenomenon originates from concentration-dependent reinforcement of the 3D network structure formed through dual hydrophobic chain association. Collectively, PLD solutions exhibit predominant elastic character.

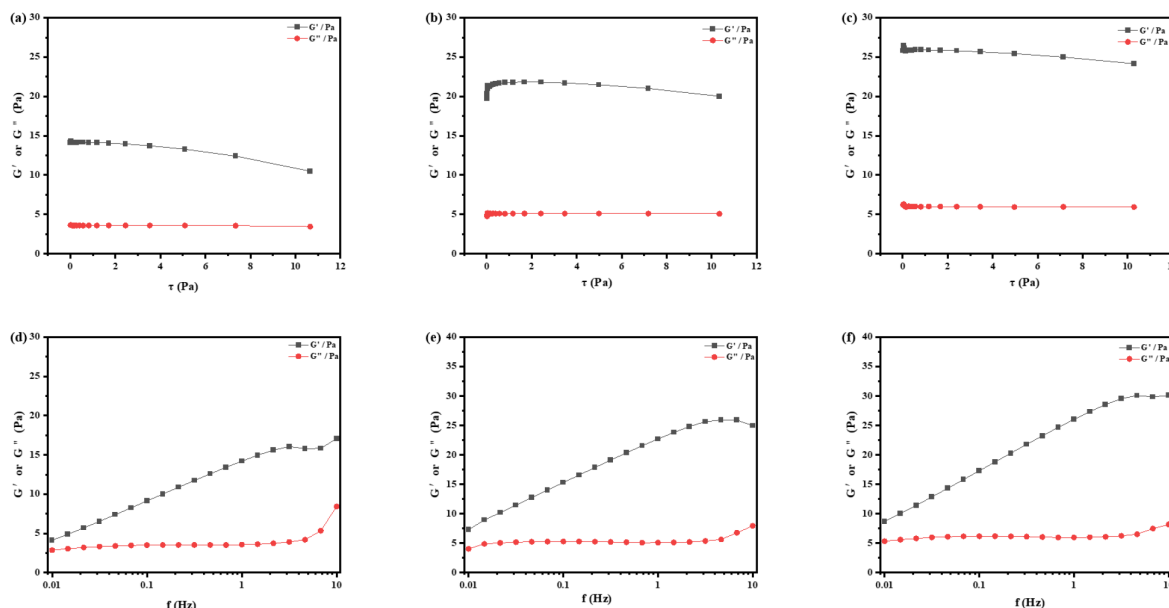


Figure 6—Viscoelastic moduli of PLD solutions: (a,d) 1.2 wt% PLD; (b,e) 1.5 wt% PLD; (c,f) 2.0 wt% PLD.

Salt Tolerance

To meet field operation requirements, NaCl and CaCl₂ were used to prepare brines with salinities of 2,500, 5,000, 10,000, 15,000, and 20,000 mg/L, simulating formation water compositions for PLD salt tolerance evaluation. As shown in Fig. 7, 1.0 wt% PLD solution viscosity progressively decreases as salinity increases from 5,000 to 20,000 mg/L. Salt ions induce salting-out effects that disrupt the polymer's solution microstructure and rheological behavior. Elevated salinity enhances electrostatic screening, causing the hydrophobic moieties on PLD chains to coil (Fig. 7a), thereby destabilizing hydrophobic microdomains, reducing hydrodynamic volume, and weakening intermolecular association. The intrinsic salt tolerance of PLD is derived from the presence of sulfonate groups in its molecular structure. Increasing the PLD concentration enhances the interchain associative network, partially offsetting the effects of salinity. Notably, 1.2 wt% PLD maintains a viscosity over 60 mPa·s after 10 min of shearing at 20,000 mg/L salinity (Fig. 7b). This demonstrates its robust salt tolerance and suitability for high-salinity reservoir applications.

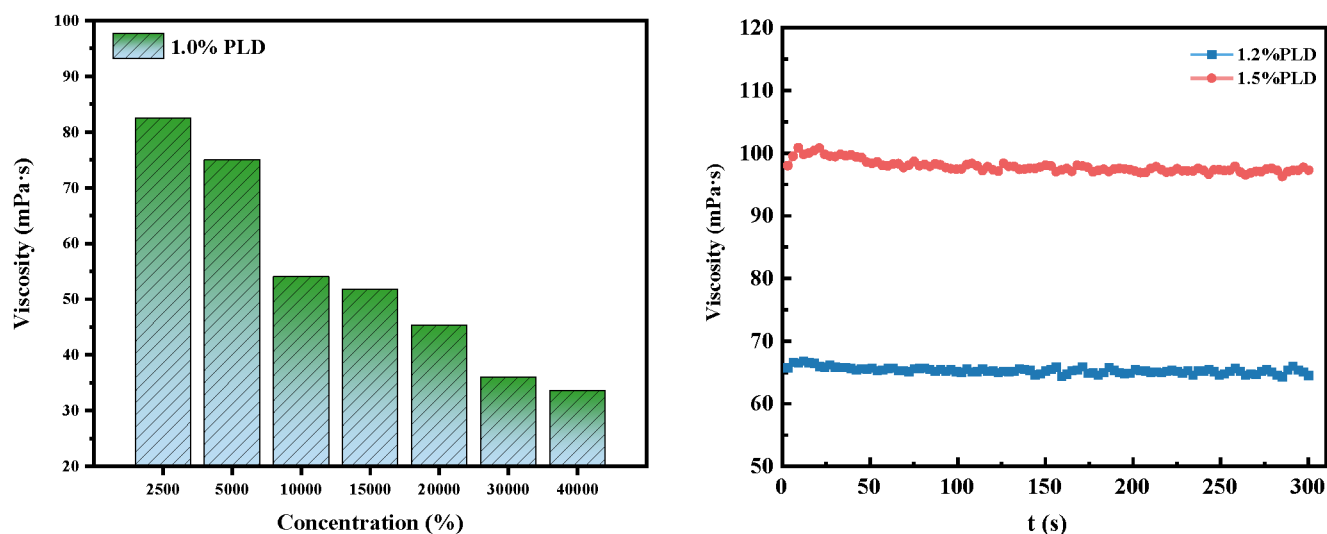


Figure 7—(a) Viscosity evolution of 1.0 wt% PLD at varying salinities;
(b) Viscosity of concentrated PLD solutions at 20,000 mg/L salinity.

Thermal and Shear Resistance

The viscosity changes of 1.2 wt% PLD from 25 °C to 120 °C and 2 wt% PLD from 25 °C to 145 °C were measured using a high-pressure sealed rheometer system. As shown in Figures 8 and 9, the apparent viscosity initially decreases with rising temperature. After reaching the target temperature, the viscosity remained above 50 mPa·s even after 90 minutes of shearing, demonstrating the good thermal and shear resistance of the PLD system. Although high temperature disrupts some of the network structure, leading to a shear-thinning response and a sharp decline in viscosity, the sulfonate groups and long-chain structure in the PLD molecular architecture contribute to its overall resistance to high temperature and shear.

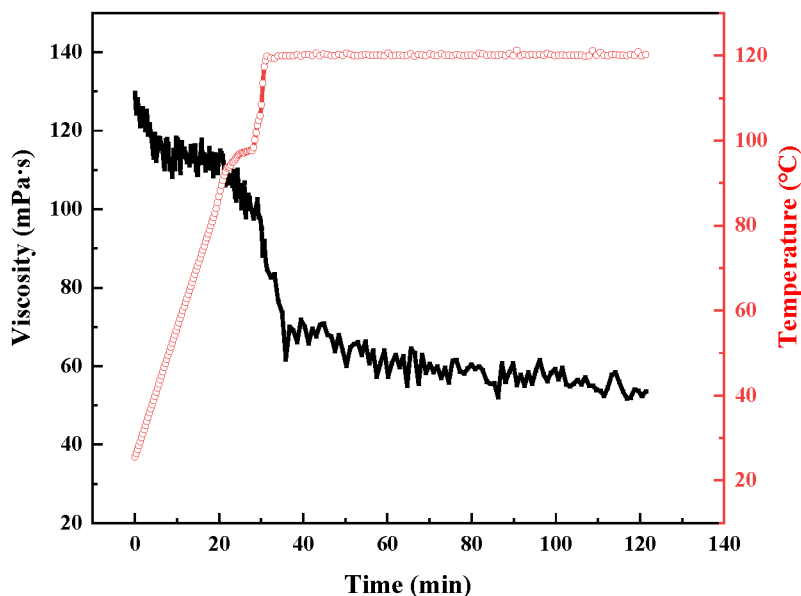


Figure 8—Evaluation of 1.2wt%PLD's thermal-shear stability.

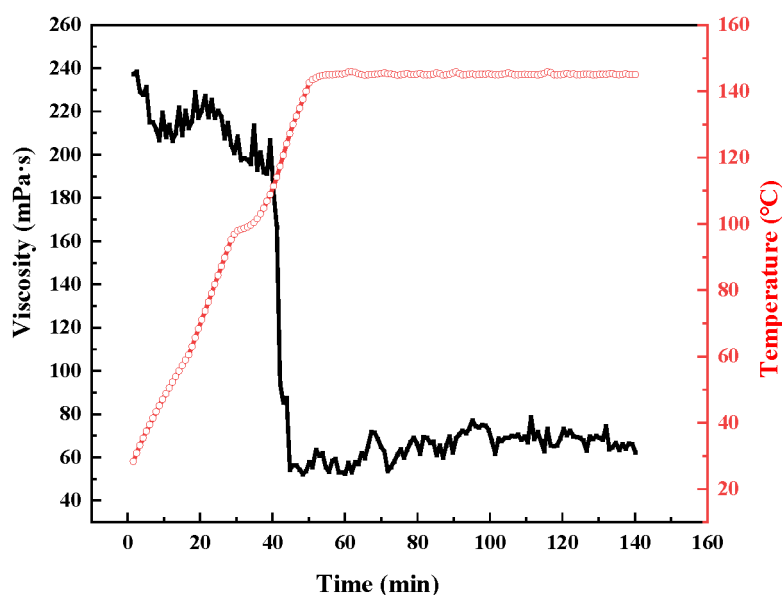


Figure 9—Evaluation of 2wt%PLD's thermal-shear stability.

Static Proppant Suspension Capacity

20/40-mesh quartz sand (30% proppant concentration) was uniformly mixed with 1.2 wt% PLD solution and 1.2% wt% hydroxypropyl guar gum (HPG) fracturing fluid, respectively. The mixtures were transferred into 50 mL graduated cylinders to monitor proppant settling rates. As quantified in Table 1, proppant settling accelerates with increasing temperature, indicating thermal degradation of network integrity. Fig. 10 demonstrates uniform proppant suspension and retarded settling in PLD solution, with significantly lower settling rates compared to HPG systems at same time. This enhanced performance originates from hydrophobic microdomain formation through intermolecular association, which establishes dense 3D networks that effectively stabilize proppant dispersion.

Table 1—Proppant settling velocities at elevated temperatures.

Fracturing Fluid	Temperature (°C)	Settling Distance (cm)	Settling Time (min)	Settling Velocity ($\text{cm} \cdot \text{min}^{-1}$)
PLD	25	5.94	122	0.049
PLD	70	6.03	78	0.077
HPG	25	6.15	97	0.063
HPG	70	6.25	54	0.116

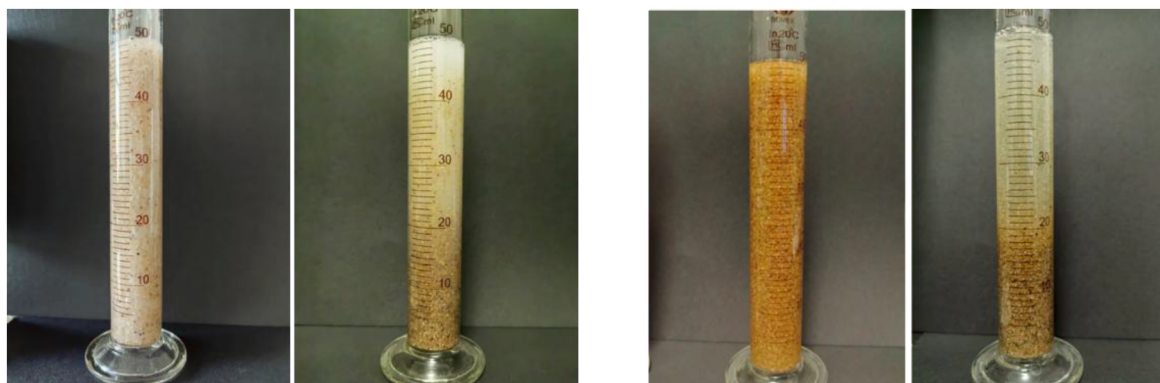


Figure 10—Proppant suspension in (a) 1.2 wt% PLD versus (b) HPG fracturing fluid.

Breaking Performance

Breakers were introduced into 0.1 wt% and 1.2 wt% PLD solutions. Sealed samples were aged in ovens at controlled temperatures to determine breaking time, broken fluid viscosity, and insoluble residue content.

As shown in Table 2, at ambient temperature with identical ammonium persulfate breaker dosage, 0.1 wt% PLD yielded lower post-break viscosity and residue than 1.2 wt% PLD. Elevated temperatures significantly shortened breaking time while progressively decreasing broken fluid viscosity and residue. Both concentrations met the post-breaking specifications for fracturing fluids.

Table 2—Breaking test results for PLD fracturing fluids.

Fracturing Fluid	Breaking Temperature (°C)	Breaker Concentration (%)	Breaking Time (min)	Broken Fluid Viscosity (mPa·s)	Insoluble Residue Content (mg·L ⁻¹)
0.1wt% PLD	25	0.1	133	2.46	2.03
1.2wt% PLD	50	0.1	153	3.06	2.59
1.2wt% PLD	70	0.1	120	2.79	2.04

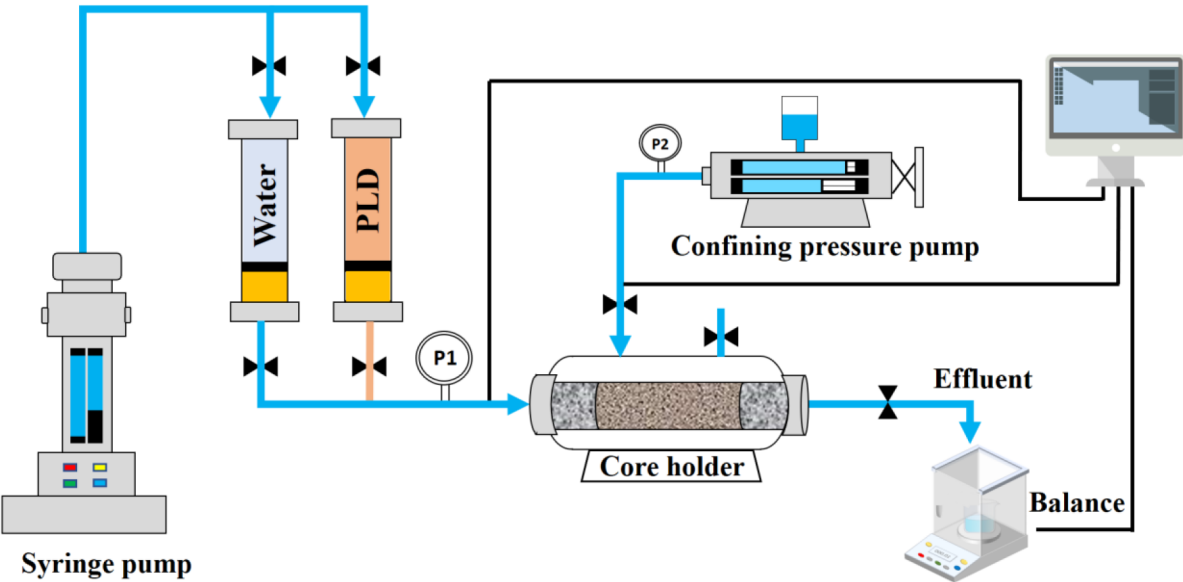


Figure 11—Schematic of the core damage experiment.

Damage

Core damage induced by 1.2% PLD spent fluid was assessed using a high-temperature/high-pressure (HTHP) core flooding system. As demonstrated in Table 3 and Fig. 11, the damage caused by PLD meets industry standards.

Table 3—Core damage test results of broken fluid.

Pre-damage Inlet Pressure (MPa)	Pre-damage Permeability (mD)	Post-damage Inlet Pressure (MPa)	Post-damage Permeability (mD)	Core Damage (%)
0.54	1.57	0.61	1.39	11.5

Conclusions

A novel drag reducer (PLD) was synthesized via copolymerization of acrylamide (AM), acrylic acid (AA), 2-acrylamido-2-methylpropanesulfonic acid (AMPS), and a dual-hydrophobic long-chain monomer (LD). Experimental results demonstrate that at low concentrations (0.1%-0.3%), PLD achieves over 70% drag

reduction in $\Phi 8$ mm pipes under a flow velocity of 12 m/s. 1.2 wt% PLD maintains exceptional stability in high-salinity environments (2×10^4 mg/L) while exhibiting favorable viscoelasticity and proppant transport capacity. 1.2% wt PLD maintain stability at temperatures up to 120°C and The broken fluid demonstrates acceptable formation damage potential, meeting operational standards for field. PLD integrates efficient drag reduction at low concentrations with robust proppant-carrying performance at high concentrations. This dual functionality meets the technical requirements for fracturing operations in the Gulong shale formation and provides an effective solution for stimulation in complex reservoirs.

Acknowledgement

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References

- Bai, Y., Chen, Y., Wu, L. et al, Research progress on nanoparticle-enhanced surfactant-based fracturing fluids: A comprehensive review[J]. *Chemical Engineering Research and Design*, 2025, **220**, 29-48.
- Cao, B.G., Chen, M.Q., Luo, P.Y. et al, Study on critical associating concentration of hydrophobic associating polymer solution[J]. *Journal of X' i an Shiyou University (Natural Science Edition)*, 2008, **23**(4): 40-42. (in Chinese)
- Deng, G., Hua, C.W., Meng, C. et al, Study on Hydrophobically Associating Cationic Polyacrylamide[J]. *Contemporary Chemical Industry*, 2021, **50**(05): 1103-1106. (in Chinese)
- He, W.Y., Wang, M., Wang, X. et al, Pore Structure Characteristics and Affecting Factors of Shale in the First Member of the Qingshankou Formation in the Gulong Sag, Songliao Basin[J]. *ACS Omega*, 2022, **7**(9), 8024-8035.
- Huang, Y.P., Hu, Y., Liu, C.L. et al, Supramolecular polymer-based gel fracturing fluid with a double network applied in ultra-deep hydraulic fracturing[J]. *Petroleum Science*, 2024, **21**(3), 1875-1888.
- Ju, C.H., Yang, E.L. Experimental Study on Stress Sensitivity Evaluation and Permeability Anisotropy of Shales in Songliao Basin[J]. *E3S Web of Conferences*, 2023, **406**, 02007.
- Lei, Z.D., Meng, S.W., Peng, Y.F. et al, Evaluation of the adaptability of CO₂ pre-fracturing to Gulong shale oil reservoirs, Songliao Basin, NE China[J]. *Petroleum Exploration and Development*, 2025, **52**(2), 459-470.
- Li, C.Y. Research on a novel polymer fracturing fluid and its application in Daqing Oilfield[J]. *Northeast Petroleum University*, 2011. (in Chinese)
- Liu, H., Huang, Y.Q., Cai, M. et al, Practice and development suggestions of hydraulic fracturing technology in the Gulong shale oil reservoirs of Songliao Basin, NE China[J]. *Petroleum Exploration and Development*, 2023, **50**(3), 688-698.
- Liu, R.Q., Shi, S.N.D, H. et al, Research on the damage of fracturing fluid to shale reservoir in Daqing[J]. *Oil Production Engineering*, 2023, **(01)**, 6-10+79-80. (in Chinese)
- Meng, S.W., Tao, J.P., Li, T.J. et al, Mechanical characteristics and reservoir stimulation mechanisms of the Gulong shale oil reservoirs, the northern Songliao Basin[J]. *Petroleum Science*, 2024, **21**(3), 2023-2036.
- Sun, L.D., Liu, H., He, W.Y., et al An analysis of major scientific problems and research paths of Gulong shale oil in Daqing Oilfield, NE China[J]. *Petroleum Exploration and Development*, 2021, **48**(3), 527-540.
- Wu, W., Liu, P.P., Wu, J.H. et al, Preparation of hydrophobically associating polymer AAMS-2 by inverse emulsion polymerization and its performance as a fracturing fluid thickener[J]. *Applied Chemical Industry*, 2016, **45**(02), 203-206. (in Chinese)
- Wu, X.L., Hou, Z.W., Yang, Y. et al, Development and performance evaluation for Gulong 1# emulsion system[J]. *Petroleum Geology & Oilfield Development in Daqing*, 2022, **41**(03), 112-119. (in Chinese)
- Xiao, D.F., Han, L., Sun, J.G. et al, Application of Polymer Fracturing Fluids in Daqing Low-Permeability Reservoirs[J]. *Oilfield Chemistry*, 2009, **26**(02), 131-134. (in Chinese)
- Yang, X., Chen, A., Mao, J. et al, Synthetic polymer fracturing fluid weighted by sodium formate enables fracture stimulations in Ultra-High pressure and High-Temperature reservoir[J]. *Fuel*, 2023, **353**, 129170.
- You, L., Xie, Y., Chen, Y. et al, Development directions of formation damage evaluation and fracturing fluids in tight and shale oil reservoirs[J]. *Petroleum*, 2025, **11**(2), 125-142.
- Yu, B., Tan, Q., Deng, J. et al, The Permeability Property and Borehole Stability in Bedding Shale[J]. *Petroleum Science and Technology*, 2013, **31**(22), 2396-2403.
- Zhang, H., Liu, H., Zheng, R., et al Suitability of utilizing alkali-activated geopolymer hollow microspheres as controlled-release carriers for breakers in gel fracturing fluids[J]. *Construction and Building Materials*, 2025, **463**, 140045.

- Zhang, Y.P., Wei, X., Tang, P.F.et al, Fracture propagating mechanism and fracturing engineering technology in Gulong shale oil reservoirs of Songliao Basin[J]. *Petroleum Geology & Oilfield Development in Daqing*, 2020, **39**(03), 170-175. (in Chinese)
- Zhang, Y.X., Pan, H.F., Bai, Y.B.et al, Pore Characteristics, Oil Contents and Factors Influencing Laminated Shale in the First Member of the Qingshankou Formation in the Gulong Sag, Northern Songliao Basin[J]. *Minerals*, 2023, **13**(9), 1220.
- Zhao, M., Yang, Z., Meng, X.et al, Enhancing temperature resistance of polymer gel fracturing fluids: The role of alcohol[J]. *Geoenergy Science and Engineering*, 2024, **242**, 213219.
- Zuo, C., Liu, P., Du, J.et al, *A review: Supramolecular polymers used in high-temperature fracturing fluids research status and prospects on the coupling relationship between molecular conformation and temperature resistance mechanism*[J]. 2025, **247**, 213699.